



## Evaluating User Satisfaction and Service Quality in Bus Rapid Transit Implementation in Amman, Jordan

Ala'a Aburomman<sup>1\*</sup>, Ala'a Alshdaifat<sup>2</sup>

<sup>1</sup> Civil Engineering Department, Al-Balqa Applied University, 19117 Salt, Jordan

<sup>2</sup> Architecture Department, The University of Jordan, 11942 Amman, Jordan

\* Correspondence: Ala'a Aburomman (alaa.aburomman@bau.edu.jo)

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**Abstract:** Since the problem of congestion in Jordanian cities is becoming more and more acute, the effectiveness of transit networks becomes critical. The current study will assess the level of user satisfaction and quality of services provided by the introduction of Bus Rapid Transit (BRT) in Amman, Jordan, to address the gap in the research on the topic after implementation. The study addresses the main questions related to the variation in rider satisfaction among the various attributes of service, changes in satisfaction and service quality after the implementation of the BRT, and also determines the priority services in the enhancement of user satisfaction. BRT users were contacted online in order to participate in a survey, in which 104 valid responses were collected through the distribution of the survey. With the findings, there is a positive perception towards accessibility, safety, and services. Also, users provide useful recommendations on how it can be improved, including the issues of queue management, expansion of routes, and improved information provision.

**Keywords:** Quality of service; User satisfaction; Bus Rapid Transit; Public transport; Travel behavior; Safety; Reliability; Vehicle characteristics

### 1 Introduction

Over the past few years, the traffic jams in Jordanian cities have become critical and unbearable. This has been greatly contributed to by the restricted accessibility and insufficiency of the passenger transport services provided by the government, which have caused most of the residents to be dependent on personal vehicles as opposed to the transport services. This has, in turn, caused great strain on the available transport systems, which require massive financial investments to reduce the issues caused by congestion. In Jordan, the major form of public passenger transport is through buses, minibuses, and shared service taxis, which operate on a fixed route basis. Better mass transit systems, such as Bus Rapid Transit (BRT) and light rail, have taken center stage, particularly in major cities such as Amman, home to more than 4 million citizens. A shift in travel behavior due to adequate mass transportation may also include the more frequent and infrequent utilization of vehicles, and the mode of transportation may also change the amount of congestion in the urban road networks [1].

Formally, the public transport in Jordan is divided into three types: buses, mini-buses, and shared taxis. The Land Transport Regulatory Commission (LTRC) [2] reported that the number of buses and mini-buses currently in operation in Jordan had reached 734 and 3,521, respectively, in 2020. Moreover, according to the Transport and Mobility Master Plan of the Greater Amman Municipality, only 13 percent of all trips to Amman were made by public transport, with 8 percent carried by taxis and 5 percent by buses [3]. Although the vehicle ownership rate is low (0.88 vehicles per 1,000 people), which is lower than the average of middle-income countries (2.66 vehicles per 1,000 people), congestion in the city is high, which points to the inefficiency of the public transport system [4, 5].

Even though the Master Plan of the Greater Amman Municipality [3] stated that only approximately 13 percent of movement in Amman was by public transport, current trends show a growing tendency to use public transport. Based on BRT system data, about 29.2 percent of all trips in Amman are now estimated to use public transport, such as BRT and buses [6]. Moreover, passenger numbers indicate that approximately 20 million journeys were

completed using Amman Bus and BRT in 2023, and this number will only increase in 2024 [7]. The satisfaction rate among the population with public transport services is also reported to be 68% as of mid-2024 [8].

Research questions are answered in this study:

1. How did rider satisfaction and perceived service quality change following the BRT implementation?
2. What dimensions of service quality do the riders feel are most important to them?
3. What are the differences in the perceptions of various dimensions of service quality as compared to general rider satisfaction?

By addressing these questions, the research aims to formulate recommendations for transportation authorities and urban planners, intending to enhance transportation services in Amman.

## **2 Literature Review**

### **2.1 User Satisfaction with Bus Rapid Transit Services**

User satisfaction has been extensively cited as one of the important indicators of BRT systems, and dimensions of service quality aspects have a significant impact on how passengers perceive them in different environments. In Latin American, South Asian, North American, and Southeast Asian studies, the importance of operational efficiency, low price, comfort, and accessibility of information is always indicated as key factors of satisfaction. According to longitudinal research conducted in Belo Horizonte, Brazil, after the introduction of BRT, its users focus on another set of priorities once transport costs, speed, reliability, and customer service have become a primary concern [9]. Likewise, the research in Multan, Pakistan, emphasizes the importance of comfort and service standards in the determination of satisfaction, stating that even systems that fulfill international standards can be characterized by moderate levels of satisfaction, and the continuous monitoring and improvement of the service are necessary [10]. Studies conducted in the United States and Indonesia also give more insight into disproportional relations between service attributes and overall satisfaction. Wu et al. [11] also found some non-linear impacts, as not all attributes, such as ease of paying the fare, had the same proportion on the degree of satisfaction, and Adriana et al. [12] observed that the BRT-Lite operations appear to be different between various urban settings, casting doubt on the sustainability of the service. The Latin American studies verify the forecasting abilities of the qualities of service quality, specifically information availability, safety, and accessibility, in the overall user satisfaction [13]. Together, these findings indicate that although satisfaction is consistently affected by the quality of services, in different situations, the importance of particular attributes may differ, which explains the importance of local and post-implementation assessment.

### **2.2 Satisfaction with Public Transportation in Jordan**

The literature on the satisfaction of people with the use of public transport has emphasized the use of conventional buses in the country instead of high-capacity transit systems. Research in secondary cities and in regional corridors suggests that secondary cities and regional corridors follow the trend of service quality, cost, reliability, and safety to affect both satisfaction and mode choice. As an example, Alomari et al. [14] revealed that the perceived quality of service is a key mediator of overall satisfaction and travel behaviour, with a focus on the enhancement of information, reliability, safety, and vehicle peculiarities. Similar reports were made by Al-Suleiman et al. [15], who evaluated public bus transport services in Jordan Valley and discussed perceived performance measures including comfort, convenience, waiting time, and safety. Nevertheless, these studies do not pay much attention to new systems of high capacity adopted in the large urban centers. There are also a few post implementation evaluations of BRT systems in the metropolitan setting (like Amman) that create a lacuna in the cognizance of real user experiences and satisfaction.

### **2.3 Study Area: Amman Bus Rapid Transit System**

The Amman BRT network is a component of the larger transport plan of the Greater Amman Municipality and runs on special lanes by a 25 km system as an alternative to personal cars, and offers high transport capacity by using the dedicated lanes. As shown in Figure 1, a total of 16 km (Corridor 1 Sweileh to Al-Mahatta) and 9 km (Corridor 2 Sports City to Ras Al-Ain) with about 100 buses serve a population of up to 142 million passengers in the projected annual ridership. The service characteristics of the Amman BRT system are summarized in Table 1. The recent literature mostly analyzes the intentions to adopt and expressed preferences of users, but not the post-implementation satisfaction. The authors used Multinational and Mixed Logit models, and 708 respondents, to determine that travel cost, waiting time, and socio-demographic factors are important determinants of willingness to use BRT [16]. Mujalli et al. [17] also noted that car ownership in households, age, gender, and being a student are factors in adoption, but with minimal general knowledge about the BRT network. These results give an idea of likely adoption and perceptions, but do not determine real satisfaction and the perceived quality of the service, which means that empirical post-implementation research should be done.



**Figure 1.** Amman Bus Rapid Transit map

**Table 1.** Amman Bus Rapid Transit service characteristics

| Service Characteristics | Line 1                                                       | Line 2                                                       |
|-------------------------|--------------------------------------------------------------|--------------------------------------------------------------|
| Path length             | 16 km                                                        | 9 km                                                         |
| Bus stops               | 21                                                           | 13                                                           |
| Travel demand           | 40 pass/hour                                                 |                                                              |
| Service time            | 16 hours (From 6 am to 10 pm)                                |                                                              |
| Service frequency       | 12 tours per daytime hour and three tours per nighttime hour | 20 daytime tours per hour and five night-time tours per hour |
| Ticket cost             | 0.55 JD                                                      | 0.55 JD                                                      |

## 2.4 Previous Research on Amman Bus Rapid Transit

Recent studies have begun to explore public perceptions and behavioral responses to the introduction of the Amman BRT system, primarily focusing on user willingness to adopt the new service and potential integration with other transport modes.

Altarifi et al. [16] employed a stated preference survey to assess users' willingness to utilize the BRT system and to evaluate demand for complementary on-demand transit services, particularly for residents located at greater distances from BRT stations. Using Multinomial Logit and Mixed Logit models based on a sample of 708 respondents, the study identified travel cost and waiting time as the most influential factors affecting mode choice. Socio-demographic characteristics, including age, income, household size, and employment status, were also found to significantly influence user preferences. The findings suggested that integrating BRT with flexible on-demand services could enhance system attractiveness, with approximately 71% of respondents expressing willingness to adopt an integrated public transport system.

Similarly, Mujalli et al. [17] investigated willingness to use the Amman BRT system through an online survey analyzed using binary logistic regression and Bayesian network models. Based on responses from 283 participants, the study found higher willingness to use BRT among households without private cars, younger users, females, and university students, particularly when fewer transfers were required. The results highlighted positive expectations regarding the system's potential to improve daily travel experiences and overall transport service quality. However, the study also revealed limited public awareness, as approximately 40% of respondents were unaware of the BRT line locations, indicating the need for enhanced information dissemination and marketing strategies.

While these studies provide valuable insights into user attitudes and adoption intentions, they primarily rely on stated preferences and expectations rather than evaluations of actual service quality and satisfaction.

## 2.5 Research Gap and Contribution

Although the literature on the evidence given on the service quality and user satisfaction in the systems of BRT is widespread in international literature, and local research examines the intentions to adopt it in Amman, there is still a substantial gap:

(1) There is little empirical research on post-implementation user satisfaction on high-capacity urban transit systems in Jordan.

(2) The available studies in the area are based on behavioral intentions or pre-implementation perceptions, but not observed user experiences.

The proposed research bridges this gap as it describes user satisfaction and perception of the service quality of the Amman BRT system upon post-implementation. Its contribution to the more complex perception of BRT performance and data-driven planning and operation strategies is through the provision of empirical evidence from a large city in the Middle East.

## 2.6 Measuring Quality of Services

umerous studies exist that investigate the correlation between transit service quality and rider satisfaction, and assist planners in understanding how the global rating of transit service is developed among the riders. Eboli and Mazzulla [18] present a mechanism of combining the passenger perceptions and measures of transit agency performance in the area of transit service quality to holistically analyze the performance of a suburban bus line. The combination of the subjective and objective indicators is expected to create a sound tool for assessing transit performance. Continuing on this, Eboli and Mazzulla [19] explore the objective measures of the transit service quality assessment and provide a comprehensive overview and interpretation of indicators. Nathanail [20] adds to this by providing a framework in which he has 22 indicators under six criteria that can be used in monitoring and controlling the quality of railway services. To increase the competitiveness of public transportation, Bozzo et al. [21] highlight other important parameters, namely the setup costs, the commercial speed, the accessibility, comfort, safety, and aesthetics. Moreover, Eboli and Mazzulla [22] use their approach in a case study of transit services in a medium-sized city, where they took into consideration such aspects as the nature of routes, reliability of the services, and environmental safety.

Based on previous research, the dimensions of service quality include Accessibility, Services, Comfort & Cleanliness, Information, and Safety and Security, as illustrated in Figure 2.



**Figure 2.** Quality of services

1. Accessibility: Path, Number of stops, and distance between stops, Bus stop location.
2. Services: Service frequency, Daily service time, Punctuality (runs that come on time), Ease of purchasing the ticket, Ticket cost, Disabled and elderly people facility.
3. Comfort & Cleanliness: Bus crowding, Comfort of seats on bus, Air conditioning on bus, Availability of shelter and benches at stop, Cleanliness of bus interior, seats, and windows.
4. Information availability: Availability of schedules/maps on the bus, Availability of schedule/maps at stops, Availability of information by phone, or mail.
5. Safety and security: Safety and competence of drivers, Security against crimes on the bus, Security against crimes at bus stops.

Although much is discussed in the existing literature regarding pre-implementation processes and the expectations of the BRT systems, there is still a significant void of information regarding the post-implementation process, especially in terms of determining the true level of satisfaction and experiences of the active users of the BRT systems. Consequently, the post-implementation phase is the area that requires additional research to gain a full picture regarding the effectiveness of the BRT system and how it is perceived by the users.

### 3 Methodology

#### 3.1 Sample Size

Mujalli et al. [17] have affirmed that Roscoe [23] suggests the determination of the necessary sample size according to the suggested guidelines that most behavioral studies should include samples greater than 30 but less than 500. The use of social media platforms to distribute the survey in this research was via Google Forms as the survey tool. Data was collected in January, 2024, and it covered 104 participants, 88.3 per cent of whom were in Amman and 11.7 per cent were out of Amman. These users were using the BRT to fulfill their transport requirements with an extension of 12 stations as shown in Figure 3, which included Swuayleh, Sports City, Al-Mahatta, the north terminal, Fountains Square, University of Jordan, the fifth circle, as well as Tareq intersection and King Abdullah gardens.



**Figure 3.** Bus Rapid Transit stations

#### 3.2 Demographic Characteristics

The first section of the questionnaire facilitated the collection of demographic information from participants. This included gender, age (in years), profession, the frequency of BRT travel per week, traveler origin, preference factors influencing BRT system choice, and place of residence, as presented in Table 2.

**Table 2.** Demographic characteristics

| Sr. No | Characteristics                                                | Categories                                 |
|--------|----------------------------------------------------------------|--------------------------------------------|
| 1      | Gender                                                         | Male/female                                |
| 2      | Age (year)                                                     | <20/21–40/41–65/>65                        |
| 3      | Profession                                                     | Student/employed/unemployed/freelance      |
| 4      | Mode used to reach the nearest Bus Rapid Transit (BRT) station | Walking/car/bus/others                     |
| 5      | Frequency of travel on BRT in a week                           | Once in a week/2–3 days/4–7 days/others    |
| 6      | Origin of traveler                                             | Work/shopping market/educational institute |
| 7      | Preference regarding BRT system due to which factor            | Schedule, time/comfort/saves money         |
| 8      | Place of living                                                | Amman/outside Amman                        |
| 9      | Station name                                                   |                                            |

#### 3.3 Likert-type Response Format

A 5-point Likert Scale would be used to assess the correspondence of the quality of the services provided by the BRT system in the city of Amman with the expectations of the users. The respondents will give subjective

assessments in 20 particular questions that will be structured into five constructs (see Table 3). The participants will be required to answer how much they agree or how much they are satisfied on a scale that will range between strongly disagree and strongly agree.

**Table 3.** Service quality dimensions

| Construct             | Number of Items |
|-----------------------|-----------------|
| Accessibility         | 3               |
| Services              | 6               |
| Comfort & Cleanliness | 5               |
| Information           | 3               |
| Safety and Security   | 3               |

### 3.4 Open-ended Question

The questionnaire incorporated two open-ended questions to delve into the overall aspects of participants' satisfaction with BRT services. The questions were structured as follows:

1. Suggest potential means or methods to enhance express bus services.
2. Share any challenges or issues you faced while using the express bus service, providing a detailed description.

### 3.5 Procedure and Measures

Research employed a combination of questionnaires and semi-structured interviews, administered to willing participants at BRT stations and through social media platforms. Data collection was conducted individually for each participant. The Excel program was used for statistical analysis.

### 3.6 Instrument Reliability and Validity

Cronbach's alpha was used to measure the internal consistency of the survey tool. All five constructs showed satisfactory reliability, with an alpha value ranging between 0.751 and 0.886, compared to the recommended alpha value of 0.70. The total scale showed a great internal consistency ( $\alpha=0.881$ ). Construct validity was assessed using exploratory factor analysis (EFA). The Kaiser Meyer Olkin (KMO) measure was 0.782, which is adequate sampling adequacy, and the Test of Sphericity was statistically significant ( $\chi^2 = 1246.3$ ,  $p = 0.001$ ). The rotated components matrix showed that all the items scored high on their intended constructs, which proved the proposed five-dimensional form of the questionnaire.

The outcomes shown in Table 4 reveal that every construct boasts of an acceptable internal consistency. The values of Cronbach's alpha are between 0.751 and 0.886, which is higher than the recommended value of 0.70. This establishes the fact that the measurement items in each construct are reliable to measure the intended dimensions of service quality and user satisfaction.

**Table 4.** Reliability analysis of measurement constructs

| Construct             | Number of Items | Cronbach's Alpha | Interpretation         |
|-----------------------|-----------------|------------------|------------------------|
| Accessibility         | 3               | 0.816            | Good reliability       |
| Services              | 6               | 0.867            | Good reliability       |
| Comfort & Cleanliness | 5               | 0.751            | Acceptable reliability |
| Information           | 3               | 0.799            | Acceptable reliability |
| Safety and Security   | 3               | 0.886            | Excellent reliability  |
| Overall scale         | 20              | 0.881            | High reliability       |

Table 5 (The KMO measure of sampling adequacy) provides the value of 0.782, which is sufficient to state that the sample size was sufficient to go through with factor analysis. The Test of Sphericity created by Bartlett was statistically significant ( $\chi^2 = 1246.3$ ,  $p = 0.001$ ), which proves that the correlation matrix is not an identity matrix and the data could be explored with the help of factor analysis.

Results of the EFA include Table 6, which indicates that all the items loaded highly on their respective dimensions. The factor loadings were between 0.61 and 0.87, which is higher than the recommended minimum of 0.50. This shows there is good convergent validity, and it corroborates the constitutive factor of the measurement model. There was sufficient representation of all the dimensions, such as Accessibility, Services, Comfort and Cleanliness, Information Availability, and Safety and Security, by items reflecting these dimensions.

**Table 5.** Kaiser Meyer Olkin (KMO) and Bartlett's test

| Test                          | Result                       |
|-------------------------------|------------------------------|
| KMO                           | 0.782                        |
| Bartlett's Test of Sphericity | $\chi^2 = 1246.3, p = 0.001$ |

**Table 6.** Exploratory factor analysis (EFA) summary

| Dimension             | Number of Items | Factor Loading Range |
|-----------------------|-----------------|----------------------|
| Accessibility         | 3               | 0.68–0.81            |
| Services              | 6               | 0.64–0.83            |
| Comfort & Cleanliness | 5               | 0.61–0.79            |
| Information           | 3               | 0.66–0.82            |
| Safety and Security   | 3               | 0.72–0.87            |

#### 4 Result

General characteristics were established through primary data collected from 104 passengers at 12 stations of the Amman BRT system. These characteristics are classified into two main groups: demographic characteristics and Quality of Services. Following each survey form, respondents were encouraged to provide suggestions for potential means or methods to enhance express bus services and describe any challenges or issues faced during their utilization of the express bus service.

##### 4.1 General Characteristics

The specific details of the general features gathered from the participants are presented in Table 7.

**Table 7.** General characteristics

| Sr. No | Characteristics                                                | Statistics                                                                                                                                         |
|--------|----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| 1      | Gender                                                         | (35.6%) Male, (64.4%) Female                                                                                                                       |
| 2      | Age (year)                                                     | (12.5%) <20, (65.4%) 21–40, (19.2%) 41–65, (2.9%) >65                                                                                              |
| 3      | Profession                                                     | (34.6%) Student, (38.5%) employed, (19.2%) unemployed, (7.7%) freelance                                                                            |
| 4      | Mode used to reach the nearest Bus Rapid Transit (BRT) station | (57.1%) Walking, (26.7%) car, (35.2%) bus, (0%) others                                                                                             |
| 5      | Frequency of travel on BRT in a week                           | (11.4%) Once a week, (21.9%) 2–3 days, (44.9%) 4–7 days, (21.8%) others                                                                            |
| 6      | Origin of traveler                                             | (31.8%) work, (27.9%) educational institute, (20.9%) shopping market, (19.4%) others                                                               |
| 7      | Preference regarding BRT system due to which factor            | (24.7%) schedule/time, (25.5%) comfort, (32%) saves money, (16.6%) saves time, (1.2%) others                                                       |
| 8      | Place of living                                                | (88.3%) Amman, (11.7%) outside Amman                                                                                                               |
| 9      | Station name                                                   | Swuayleh; Sports City; Al-Mahatta; North Terminal; Fountains Square; University of Jordan; Fifth Circle; Tareq Intersection; King Abdullah Gardens |

A diverse makeup of the participants in the research study is demonstrated by the demographic profile of the respondents, where (35.6) were male and (64.4) were female. Age structure is as follows: (12.5) below 20 years, (65.4) between 21 and 40 years, (19.2) between 41 and 65 years, and (2.9) above 65 years. The professions are diverse, and (34.6%) students, (38.5) employed people, (19.2) not employed, and (7.7) freelancers. When it comes to the methods of getting to the closest BRT station, a great percentage (57.1%) will walk, (26.7%) will drive, (35.2%) will take a bus, and no one indicated other possibilities. The frequency of BRT traveling is (11.4) 1 time per week, (21.9) 2–3 days, (44.9) 4–7 days, and (21.8) 0–21.8 days. The purpose of traveling differs: (31.8) is work purpose, (27.9) is education purpose, and (20.9) is shopping purpose. The BRT system or preference revolves around such aspects as (24.7) schedule and time, (25.5) comfort, (32) cost-effectiveness, and (16.6) time efficiency. Most of them (88.3%) are found in Amman, and the participants use a variety of BRT stations, such as Swuayleh, Sports City,

Al-Mahatta, North Terminal, Fountains Square, University of Jordan, the Fifth Circle, Tareq Intersection, and King Abdullah Gardens.

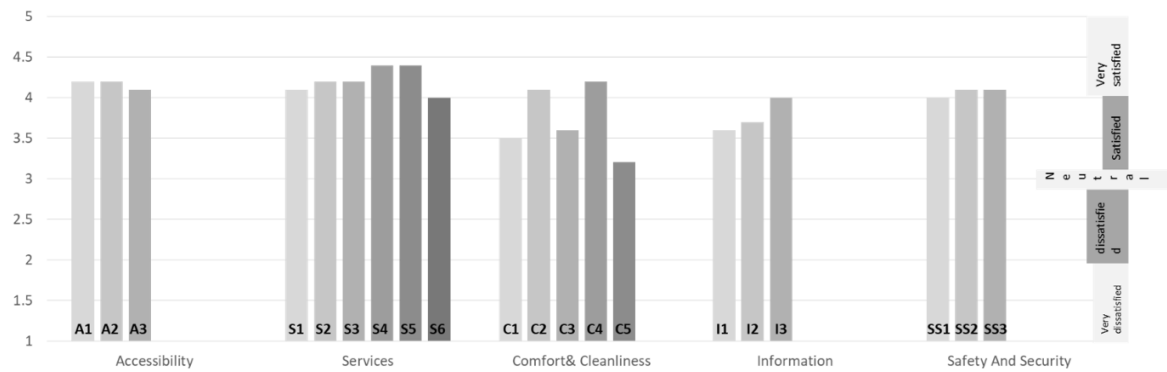
## 4.2 Quality of Services with Amman Bus Rapid Transit

To gauge respondents' subjective evaluations, users' perceptions of BRT utilization were assessed through 20 specific questions organized into five constructs, as outlined in Table 8.

**Table 8.** Questionnaire survey statistics

| Construct             |     | Items                                                                                  | Numerical Value |    |    |    |    |       |
|-----------------------|-----|----------------------------------------------------------------------------------------|-----------------|----|----|----|----|-------|
|                       |     |                                                                                        | 1               | 2  | 3  | 4  | 5  | Total |
| Accessibility         | A1  | The path of the bus is clear and easy to understand.                                   | 2               | 2  | 15 | 35 | 50 | 4.2   |
|                       | A2  | The number of stops and the distance between stops are convenient for my travel needs. | 1               | 8  | 10 | 29 | 56 | 4.2   |
|                       | A3  | Bus stop locations are easily accessible.                                              | 3               | 8  | 13 | 26 | 54 | 4.1   |
| Services              | S1  | The frequency of the bus service meets my expectations.                                | 6               | 6  | 14 | 17 | 61 | 4.1   |
|                       | S2  | The daily service times are suitable for my commuting schedule.                        | 6               | 3  | 11 | 18 | 66 | 4.2   |
|                       | S3  | The bus service is punctual.                                                           | 5               | 2  | 12 | 24 | 61 | 4.2   |
|                       | S4  | Purchasing a ticket is easy.                                                           | 7               | 0  | 10 | 12 | 75 | 4.4   |
|                       | S5  | The ticket cost is reasonable.                                                         | 3               | 0  | 14 | 19 | 68 | 4.4   |
|                       | S6  | The bus service is accommodating for individuals with disabilities.                    | 6               | 7  | 15 | 22 | 54 | 4.0   |
| Comfort & Cleanliness | C1  | Bus crowding is a concern during my travels.                                           | 13              | 16 | 15 | 24 | 36 | 3.5   |
|                       | C2  | The seats on the bus are comfortable.                                                  | 2               | 7  | 12 | 36 | 47 | 4.1   |
|                       | C3  | Satisfaction with air conditioning.                                                    | 23              | 2  | 9  | 25 | 45 | 3.6   |
|                       | C4  | Bus stops have adequate shelter and benches.                                           | 4               | 1  | 11 | 34 | 54 | 4.2   |
|                       | C5  | The cleanliness of the bus interior, including seats and windows, is satisfactory.     | 25              | 9  | 18 | 18 | 34 | 3.2   |
| Information           | I1  | Schedule/maps on the bus and announcements are readily available.                      | 8               | 16 | 15 | 26 | 39 | 3.6   |
|                       | I2  | Schedule/maps at bus stops are easily accessible.                                      | 8               | 17 | 13 | 25 | 41 | 3.7   |
|                       | I3  | Information is easily available by phone, mail, or other channels.                     | 6               | 7  | 14 | 27 | 50 | 4.0   |
| Safety and Security   | SS1 | The drivers demonstrate safety and competence.                                         | 6               | 10 | 9  | 28 | 51 | 4.0   |
|                       | SS2 | I feel secure against crimes while on the bus.                                         | 6               | 5  | 14 | 22 | 57 | 4.1   |
|                       | SS3 | I feel secure against crimes at bus stops.                                             | 6               | 2  | 16 | 30 | 50 | 4.1   |

The survey items were assessed on a scale ranging from 1 to 5, with 1 indicating low satisfaction and 5 indicating high satisfaction. Under the Accessibility category, respondents expressed positive views, with A1 (The path of the bus is clear and easy to understand) and A2 (The number of stops and distance between stops is convenient for my travel needs) both receiving a rating of 4.2, while A3 (Bus stop locations are easily accessible) received a rating of 4.1. In the Services category, S4 (Purchasing a ticket is easy) and S5 (The ticket cost is reasonable) both received the highest rating of 4.4, indicating a strong positive sentiment. Comfort & Cleanliness items showed varying levels of satisfaction, with C2 (The seats on the bus are comfortable) and C4 (Bus stops have adequate shelter and benches) receiving ratings of 4.1 and 4.2, respectively, while C5 (The cleanliness of the bus interior, including seats and windows, is satisfactory) had a lower rating of 3.2. In the Information category, respondents were generally satisfied, with I3 (Information is easily available by phone, mail, or other channels) receiving a rating of 4.0. Finally, in the Safety and Security category, all items scored well, with SS1 (The drivers demonstrate safety and competence), SS2



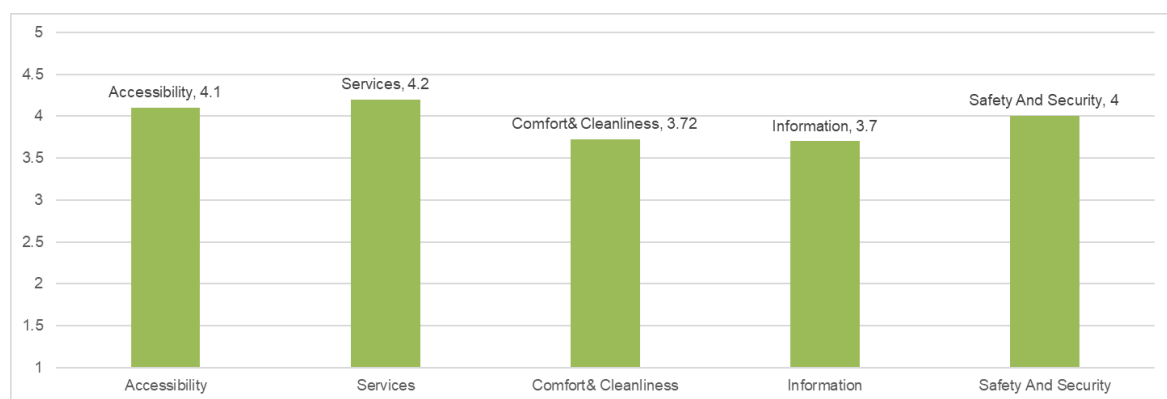
**Figure 4.** Questionnaire survey statistics

(I feel secure against crimes while on the bus), and SS3 (I feel secure against crimes at bus stops) receiving ratings of 4.0, 4.1, and 4.1, respectively as shown in Figure 4.

Table 9 on the overall satisfaction scores on the various categories in the survey indicated the following: There was a rating of 4.1 on Accessibility, which means that there is a positive perception about the ease and convenience of bus travel. The rating of services was 4.2, which indicates a high degree of satisfaction with the aspects of purchasing tickets, the frequency of services, and the time at which services are provided. Comfort & Cleanliness received a rating of 3.72, which implies that they were moderately satisfied with the features of the bus comfort and cleanliness. The score of information was 3.7, which showed that there is an overall positive response towards the availability of information through different channels. The rating of Safety and Security was 4, and it reflects a strong sense of safety and competence, both related to bus travel and bus stops. The total satisfaction score was 3.9 in all categories, indicating that there is a general positive attitude of the respondents towards their experiences with the bus services.

**Table 9.** Satisfaction scores

| Construct             | Total |
|-----------------------|-------|
| Accessibility         | 4.1   |
| Services              | 4.2   |
| Comfort & Cleanliness | 3.72  |
| Information           | 3.7   |
| Safety and Security   | 4.0   |
| Total                 | 3.9   |



**Figure 5.** Satisfaction scores

As shown in Figure 5, the mean rating of 3.9 was achieved when the arithmetic mean of the ratings was calculated in all five dimensions of service quality. To summarize the perceptions of the users and give a general overview of the level of satisfaction, descriptive statistics, such as the mean values and standard deviation, were calculated with the help of Microsoft Excel. The analysis of the responses to each service quality dimension was conducted with the help of the identical descriptive measures. The total score, which is the sum of the five dimension scores divided by the average score, is a general measure of how the users perceive the BRT service.

The reliability analysis confirmed that all measurement constructs exhibit acceptable to excellent internal consistency. Safety and security achieved the highest reliability, while comfort, and cleanliness showed adequate consistency.

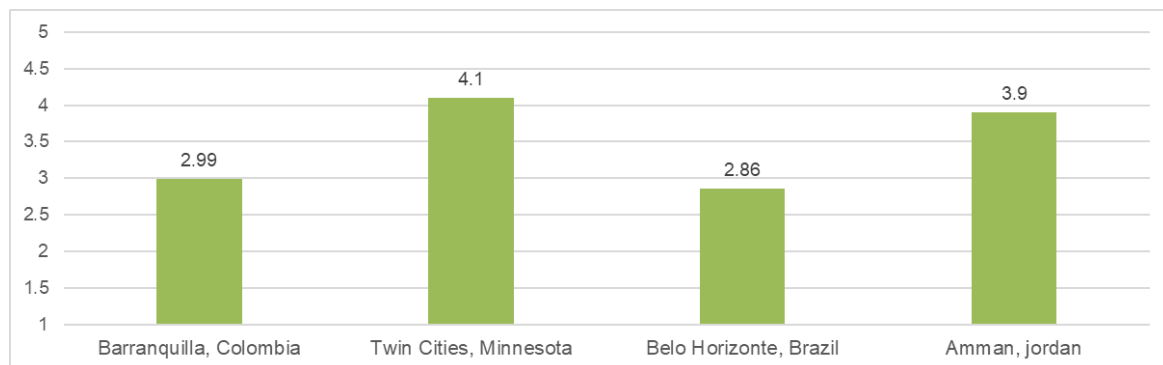
The EFA validated the conceptual framework of the study, as all questionnaire items loaded significantly onto their respective dimensions. These findings confirm that the survey instrument is both reliable and valid for assessing user satisfaction with the Amman BRT system.

## 5 Limitations

Despite the fact that the sample is large enough to approve the minimum standards of exploratory behavioral research, the research has some limitations as concerns the sample representativeness. The analysis will be done using 104 valid responses, which might not be a complete representation of the diversity in terms of travel behavior, socio-economic features and mobility of the people of Amman, which have more than four million people as residents. In connection with this, the results can be viewed as indicative, but not entirely applicable. However, the sample size is deemed to be sufficient to reflect the initial user perception in post implementation evaluation studies especially at the initial stages of operation of new transit systems. It is suggested that the future studies should use bigger and more varied samples to obtain a better representativeness and allow wider generalization of the findings.

## 6 Discussion

The BRT system in Amman was observed to result in passenger satisfaction. This satisfaction implies that the ease of usage of the system was generally rated in a positive manner, which was a successful transition to the traditional mode of transport. Such a change is crucial in altering the perception of people and making the rapid bus system among the viable and desirable choices of the passengers in Amman. A comparison between the quality of services offered within the BRT system in Amman and the results of the prior studies carried out in different global settings provides a holistic vision of the efficiency and issues related to the BRT application. As Figure 6 illustrates, it is quite clear that the city of Amman has scored a good general satisfaction rate, as it is observed relative to other cities.

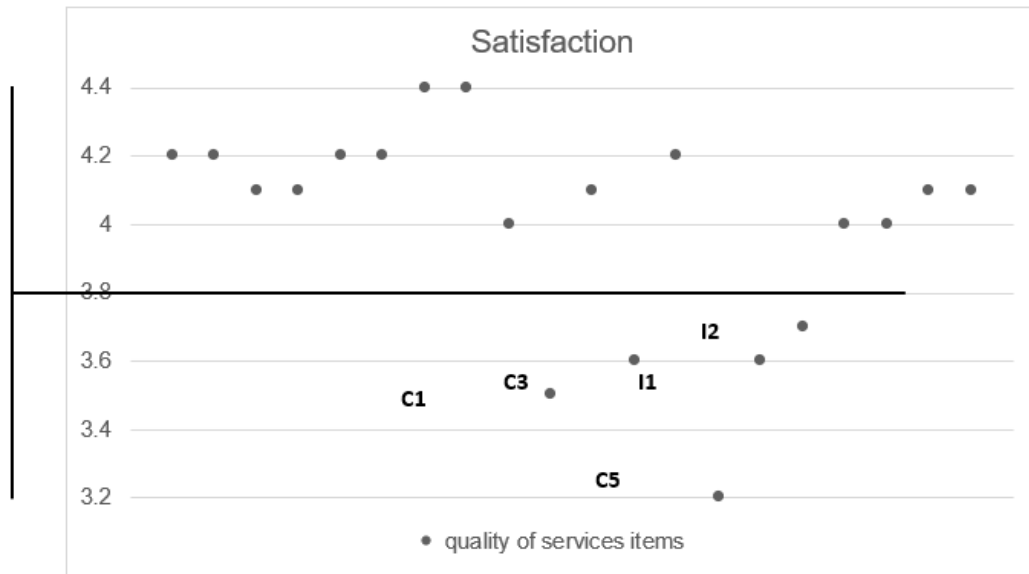


**Figure 6.** Comparison with previous studies

Major influential services can be identified to provide good information on the most important areas that lead to a substantial contribution to rider satisfaction. We plotted the 20 attributes on a chart, as it would show where they were performing poorly and needed improvement, and their average performance was plotted on the y-axis (Figure 7). The following visual tool is utilized in determining the service attributes that could be improved. The focus should be on the elements where performance is low, as they are the fundamentals of the overall satisfaction levels, and can be improved. The total balance of service attributes gave a performance that was above 3.8, which is the reference level. Generally speaking, not a single characteristic that was rated on a scale demonstrates that there is a high level of dissatisfaction among the passengers. Nonetheless, the other attributes got a relatively low rating, and this shows that there can be improvement. There were five characteristics that were lower than the mean bus crowding, air conditioning in the bus, seat and interior window cleanliness, timetables/map availability in the bus, and timetables/map availability in the stations. These findings allow us to determine some spheres where the impressions of the users are not so favorable and provide guidance on what to work on in the further enhancement of the service.

The results of this study provide several important insights that can inform transportation planning and policy development. Based on the key findings, a set of policy recommendations has been developed to enhance the effectiveness and quality of the public transport system. These recommendations focus on improving service reliability, accessibility, passenger satisfaction, and operational efficiency. As summarized in Table 10, the proposed

policies aim to address the main issues identified through the analysis and provide practical guidance for decision-makers and transport authorities to improve the overall performance and sustainability of the system.



**Figure 7.** Service quality components

Note: C1—Bus crowding is a concern during my travels; C3—Satisfaction with air conditioning; C5—The cleanliness of the bus interior, including seats and windows, is satisfactory; I1—Schedule/maps on the bus and announcements are readily available; and I2—Schedule/maps at bus stops are easily accessible.

**Table 10.** Policy recommendations

| Services                                      | Code | Max | Min | Satisfaction |
|-----------------------------------------------|------|-----|-----|--------------|
| Path                                          | A1   | 5   | 1   | 4.2          |
| Number of stops and distance between stops    | A2   | 5   | 1   | 4.2          |
| Bus stop location                             | A3   | 5   | 1   | 4.1          |
| Service frequency                             | S1   | 5   | 1   | 4.1          |
| Daily service time                            | S2   | 5   | 1   | 4.2          |
| Punctuality (runs that come on time)          | S3   | 5   | 1   | 4.2          |
| Ticket cost                                   | S4   | 5   | 1   | 4.4          |
| The ticket cost is reasonable.                | S5   | 5   | 1   | 4.4          |
| Disabled and elderly people facility          | S6   | 5   | 1   | 4.0          |
| Bus crowding                                  | C1   | 5   | 1   | 3.5          |
| Comfort of seats on bus                       | C2   | 5   | 1   | 4.1          |
| Air conditioning on bus                       | C3   | 5   | 1   | 3.6          |
| Availability of shelter and benches at stop   | C4   | 5   | 1   | 4.2          |
| Cleanliness of bus interior seats and windows | C5   | 5   | 1   | 3.2          |
| Availability of schedule/maps on bus          | I1   | 5   | 1   | 3.6          |
| Availability of schedule/maps at stops        | I2   | 5   | 1   | 3.7          |
| Availability of information by phone, mail    | I3   | 5   | 1   | 4.0          |
| Safety and competence of drivers              | SS1  | 5   | 1   | 4.0          |
| Security against crimes on bus                | SS2  | 5   | 1   | 4.1          |
| Security against crimes at bus stops          | SS3  | 5   | 1   | 4.1          |

## 7 Conclusion

The open-ended questions were analyzed through the thematic analysis method to determine the problems and areas of improvement that could be made to the BRT system in Amman. Overall, the BRT system was positively perceived and, in particular, its effectiveness and reliability in addressing the transportation demand on a daily basis. The qualitative analysis identified four major themes. The former theme is linked to the station management, where

some respondents recorded the need to have certain physical barriers to increase control over the queue of boarding and alighting at all the stations. One of the participants wrote, “Without physical obstacles, queues at peak times would be disorderly, and boarding and alighting would not be organized (Participant 83). The second theme is expansion of the network, in which the participants opined that the BRT services should have been expanded to other places to make the area more accessible, due to the comments by one of the respondents, who stated that extension of the BRT service to the adjacent cities would greatly enhance daily commuting opportunities (Participant 16). The third theme is the provision of information and wayfinding, which is essential to have clear explanations of the trip within buses, and brochures and maps of all express and feeder routes. One of the participants commented that there should be maps and brochures in the buses, which will familiarize them with all routes and connections (Participant 5). The final theme is related to the comfort of the passengers, and some of the customers recommend installing interior blinds to be distracted by the unpleasant inconveniences in the journey. One of the users said that the blinds should be installed inside the buses to shield the passengers against direct sunlight, mostly during summertime (Participant 22). The direct quotes serve to elucidate the identified themes and provide a supplement to the quantitative findings, since they will provide a deeper understanding of the experiences and expectations of the users. The recommendations provided by the users can be considered in future planning and operational decision-making in order to define how the quality of the services can be enhanced to help continue the usage of the BRT system in Amman and make it sustainable.

According to the research results that assess the user satisfaction with BRT services, the following are improvements that are recommended:

1. **Enhance Crowding Management:** Solve the issue of bus crowding through controlling the flow of passengers. Introduce (occupancy monitoring through mobile applications in real time), implement (crowd management staff during rush hours), and think over (digital notifications) to educate passengers about the bus capacity. The actions will curb congestion and enhance the general customer experience.

2. **Improve Bus Interior Conditions:** Improve sanitation through developing routines of cleaning seats and windows. Introduction of onboard sanitization stations and regular inspections to keep the hygiene at the required level. This will establish a user-friendly and secure atmosphere, which will have a positive impact on passenger satisfaction.

3. **Maximize Air Conditioning Systems:** The air conditioning should be maintained in a proper condition by performing regular repairs and upgrades on time. Manage the effectiveness of the monitoring system and make use of the automated sensors to create comfort in the hot weather to prevent situations where passengers feel uncomfortable and ensure their satisfaction.

4. **Increase Availability of Information:** Increase bus and station schedules. Install electronic screens or interactive screens on the buses that demonstrate real-time route information and schedule. Ensure that revised schedules/maps are appropriately positioned in all the stations to ensure that one navigates without any hustle.

5. **Install Queue Management:** In any station, install barriers and automated gates that have sensors to control boarding and exiting. To it, add-on smart queue monitoring systems to improve orderliness, not to mention operational efficiency during peak.

6. **Give Full Trip Information:** Ensure that the passengers know about the future stops and transfers by giving detailed descriptions of each trip on the onboard digital screens and mobile notifications. Add them to the utilization of the real-time updates to enhance the traveling experience.

7. **Install Sun Shields (Blinds):** Install versatile and inexpensive blinds to protect the passengers against the direct sunlight. Ensure that the design allows the passengers to control their exposure to light themselves, which boosts comfort in travel.

8. **Frequent User Feedback Mechanism:** Have a regular mechanism for gathering and reviewing user feedback. Proactively use user input to make decisions to ensure that the services are constantly improved according to the growing needs and preferences of the passengers.

Overall, the findings suggest that the Amman BRT system is perceived by passengers as efficient, reliable, and responsive. The rating of most service attributes was high beyond the good level, whereas some attributes, e.g., bus crowding, air conditioning, cleanliness, and availability of timetables/maps, were relatively low. Quantitative rating, along with the qualitative feedback, gives a full picture of user experiences and identifies particular dimensions, the perception of which is not as positive. This is because these results can provide descriptive knowledge that may be utilized to enhance monitoring in the future, as well as planning and possible enhancements in the BRT system in accordance with the expectations of the users in the Amman urban transport setting.

## Author Contributions

Conceptualization, A.Aburomman and A.Alshdaifat; methodology, A.Aburomman and A.Alshdaifat; formal analysis, A.Aburomman and A.Alshdaifat; investigation, A.Aburomman; data curation, A.Aburomman; writing—original draft preparation, A.Aburomman; writing—review and editing, A.Aburomman; supervision, A.Aburomman; project

administration, A.Aburomman; funding acquisition, A.Aburomman. All authors have read and agreed to the published version of the manuscript.

### Data Availability

The data used to support the findings of this study are available from the corresponding author upon request.

### Conflicts of Interest

The authors declare that they have no conflicts of interest.

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